

```
pip install numpy
```

Defaulting to user installation because normal site-packages is not writeable

Collecting numpy

Using cached numpy-2.0.2-cp39-cp39-macosx_14_0_arm64.whl (5.3 MB)

Installing collected packages: numpy

Successfully installed numpy-2.0.2

WARNING: You are using pip version 21.2.4; however, version 25.0.1 is available.

You should consider upgrading via the

'/Library/Developer/CommandLineTools/usr/bin/python3 -m pip install --upgrade pip' command.

Note: you may need to restart the kernel to use updated packages.

```
pip install pandas matplotlib seaborn
```

Defaulting to user installation because normal site-packages is not writeable

Collecting pandas

Using cached pandas-2.2.3-cp39-cp39-macosx_11_0_arm64.whl (11.3 MB)

Collecting matplotlib

Using cached matplotlib-3.9.4-cp39-cp39-macosx_11_0_arm64.whl (7.8 MB)

Collecting seaborn

Using cached seaborn-0.13.2-py3-none-any.whl (294 kB)

Collecting tzdata>=2022.7

Downloading tzdata-2025.2-py2.py3-none-any.whl (347 kB)

Requirement already satisfied: numpy>=1.22.4 in

/Users/jagdeepsingh/Library/Python/3.9/lib/python/site-packages (from pandas) (2.0.2)

Requirement already satisfied: python-dateutil>=2.8.2 in

/Users/jagdeepsingh/Library/Python/3.9/lib/python/site-packages (from pandas) (2.9.0.post0)

Collecting pyparsing>=2.3.1

Downloading pyparsing-3.2.3-py3-none-any.whl (111 kB)

Requirement already satisfied: packaging>=20.0 in

/Users/jagdeepsingh/Library/Python/3.9/lib/python/site-packages (from matplotlib) (24.2)

Collecting kiwisolver>=1.3.1

Using cached kiwisolver-1.4.7-cp39-cp39-macosx_11_0_arm64.whl (64 kB)

Collecting cyclor>=0.10

Using cached cyclor-0.12.1-py3-none-any.whl (8.3 kB)

Collecting pillow>=8

Downloading pillow-11.2.1-cp39-cp39-macosx_11_0_arm64.whl (3.0 MB)

portlib-resources>=3.2.0

Using cached importlib_resources-6.5.2-py3-none-any.whl (37 kB)

Collecting fonttools>=4.22.0

Downloading fonttools-4.57.0-cp39-cp39-macosx_10_9_universal2.whl

```
(2.8 MB)
acosx_11_0_arm64.whl (249 kB)
Requirement already satisfied: zipp>=3.1.0 in
/Users/jagdeepsingh/Library/Python/3.9/lib/python/site-packages (from
importlib-resources>=3.2.0->matplotlib) (3.21.0)
Requirement already satisfied: six>=1.5 in
/Library/Developer/CommandLineTools/Library/Frameworks/Python3.framework/Versions/3.9/lib/python3.9/site-packages (from python-
dateutil>=2.8.2->pandas) (1.15.0)
Installing collected packages: tzdata, pytz, pyparsing, pillow,
kiwisolver, importlib-resources, fonttools, cyclor, contourpy, pandas,
matplotlib, seaborn
Successfully installed contourpy-1.3.0 cyclor-0.12.1 fonttools-4.57.0
importlib-resources-6.5.2 kiwisolver-1.4.7 matplotlib-3.9.4 pandas-
2.2.3 pillow-11.2.1 pyparsing-3.2.3 pytz-2025.2 seaborn-0.13.2 tzdata-
2025.2
WARNING: You are using pip version 21.2.4; however, version 25.0.1 is
available.
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Note: you may need to restart the kernel to use updated packages.
```

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from mpl_toolkits.mplot3d import Axes3D
%matplotlib inline
```

Matplotlib is building the font cache; this may take a moment.

```
data=pd.read_csv("/Users/jagdeepsingh/Downloads/Mall_Customers.csv")
data.head()
```

	CustomerID	Gender	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

```
data['Gender'] = data['Gender'].map({'Male': 0, 'Female': 1}) (#
Convert 'Gender' column to numeric values: Male = 0, Female = 1, for
correlation or ML models)
)
data.corr()
```

	CustomerID	Gender	Age	Annual Income
(k\$) \				
CustomerID	1.000000	-0.057400	-0.026763	
0.977548				
Gender	-0.057400	1.000000	-0.060867	-
0.056410				
Age	-0.026763	-0.060867	1.000000	-
0.012398				
Annual Income (k\$)	0.977548	-0.056410	-0.012398	
1.000000				
Spending Score (1-100)	0.013835	0.058109	-0.327227	
0.009903				

	Spending Score (1-100)
CustomerID	0.013835
Gender	0.058109
Age	-0.327227
Annual Income (k\$)	0.009903
Spending Score (1-100)	1.000000

```
plt.figure(figsize=(10, 6))
sns.set(style = 'whitegrid')
sns.distplot(data['Annual Income (k$)'])
plt.title('Distribution of Annual Income (k$)', fontsize = 20)
plt.xlabel('Range of Annual Income (k$)')
plt.ylabel('Count')
```

/var/folders/22/ns7lfhrj1w70915sgd7n8wkm0000gn/T/
ipykernel_51292/2958858929.py:3: UserWarning:

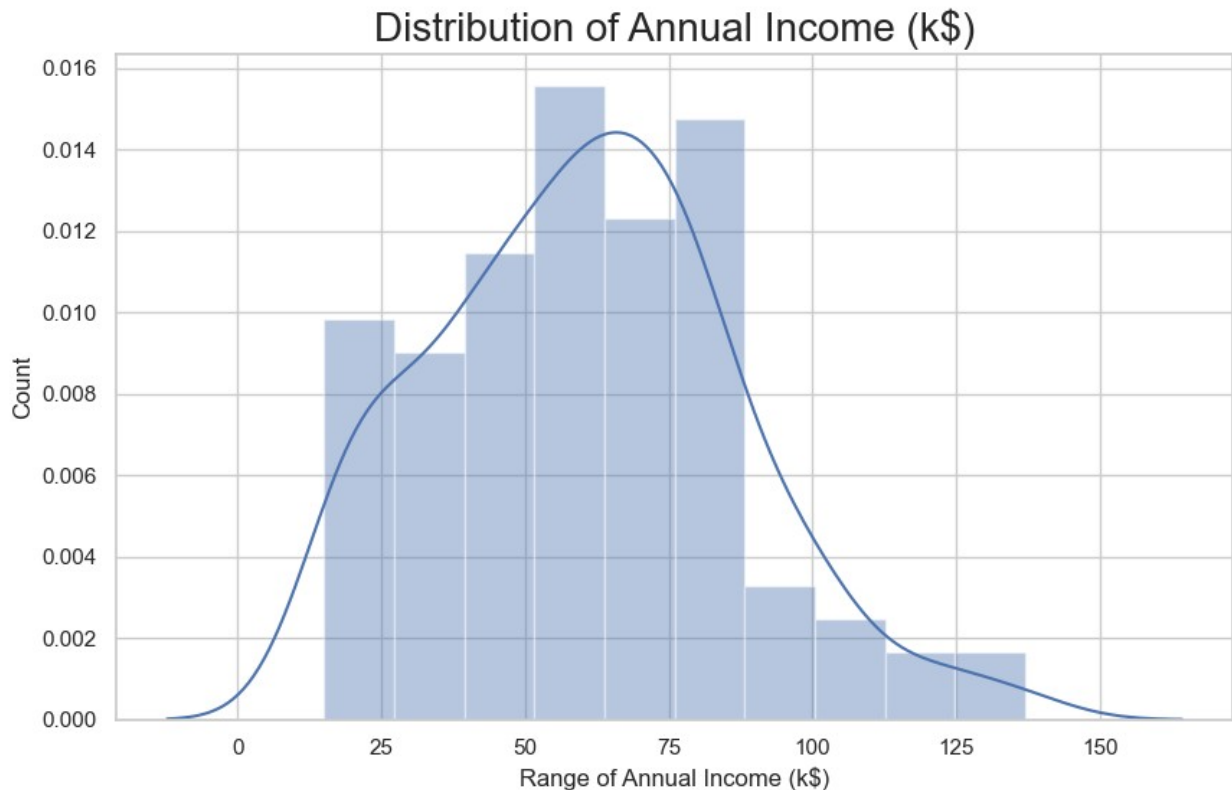
`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see <https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751>

```
sns.distplot(data['Annual Income (k$)'])
```

```
Text(0, 0.5, 'Count')
```



```
#Distribution of age
plt.figure(figsize=(10, 6))
sns.set(style = 'whitegrid')
sns.distplot(data['Age'])
plt.title('Distribution of Age', fontsize = 20)
plt.xlabel('Range of Age')
plt.ylabel('Count')
```

/var/folders/22/ns7lfrhj1w70915sgd7n8wkm0000gn/T/
ipykernel_51292/3692109745.py:4: UserWarning:

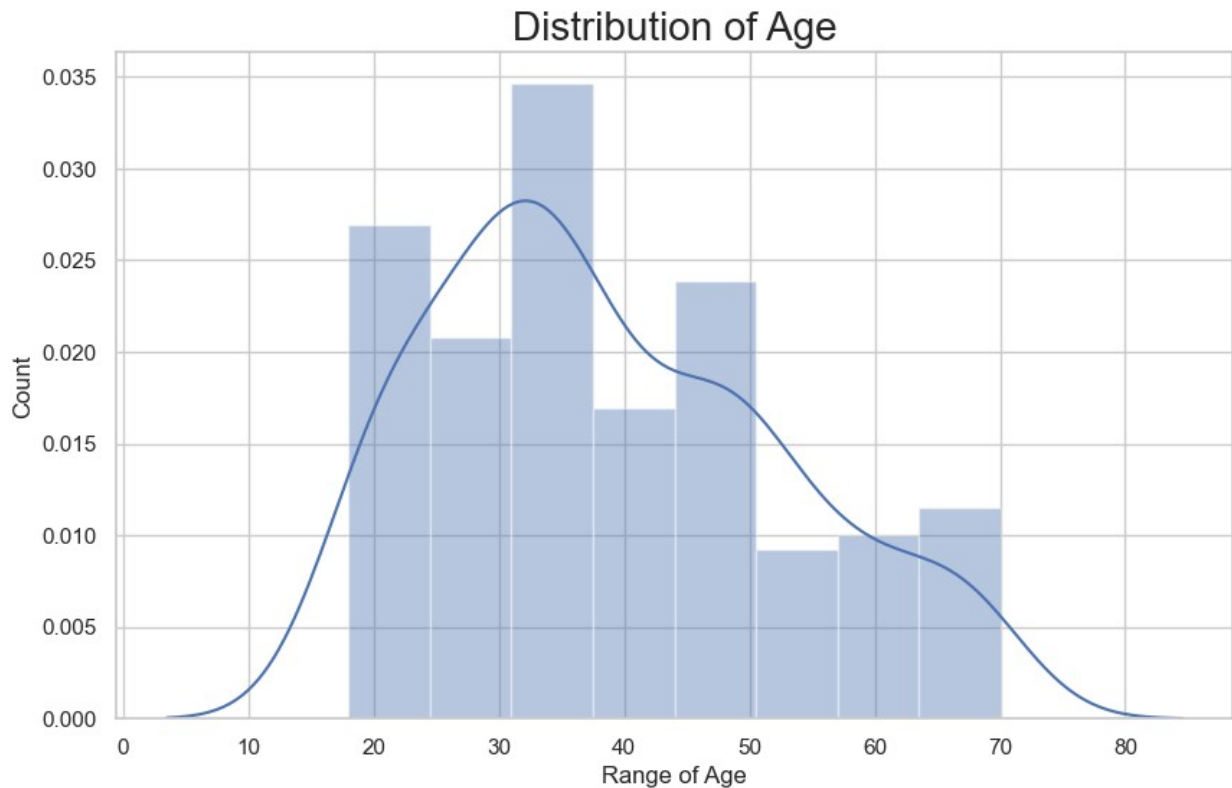
`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see <https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751>

```
sns.distplot(data['Age'])
```

```
Text(0, 0.5, 'Count')
```



```
#Distribution of spending score
plt.figure(figsize=(10, 6))
sns.set(style = 'whitegrid')
sns.distplot(data['Spending Score (1-100)'])
plt.title('Distribution of Spending Score (1-100)', fontsize = 20)
plt.xlabel('Range of Spending Score (1-100)')
plt.ylabel('Count')
```

/var/folders/22/ns7lfrhjlw70915sgd7n8wkm0000gn/T/
ipykernel_51292/2493241498.py:4: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

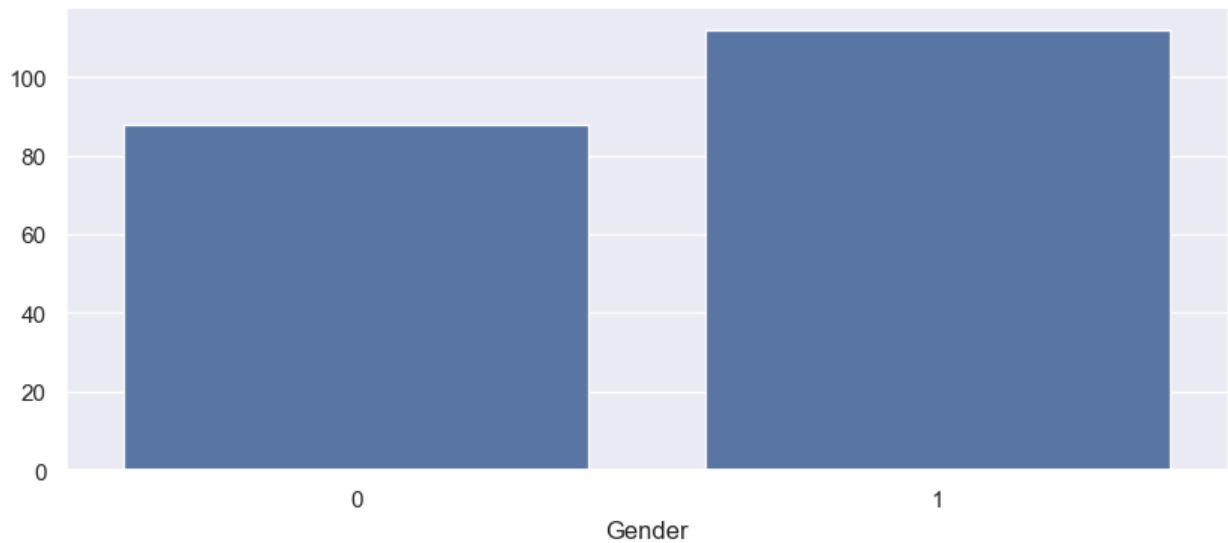
Please adapt your code to use either `displot` (a figure-level function with similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see <https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751>

```
sns.distplot(data['Spending Score (1-100)'])
Text(0, 0.5, 'Count')
```



```
genders = data.Gender.value_counts()  
sns.set_style("darkgrid")  
plt.figure(figsize=(10,4))  
sns.barplot(x=genders.index, y=genders.values)  
plt.show()
```



```
df1=data[["CustomerID","Gender","Age","Annual Income (k$)","Spending Score (1-100)"]]
X=df1[["Annual Income (k$)","Spending Score (1-100)"]]
```

```
X.head()
```

	Annual Income (k\$)	Spending Score (1-100)
0	15	39
1	15	81
2	16	6
3	16	77
4	17	40

```
#Scatterplot of the input data
```

```
plt.figure(figsize=(10,6))
sns.scatterplot(x = 'Annual Income (k$)',y = 'Spending Score (1-100)',
data = X ,s = 60 )
plt.xlabel('Annual Income (k$)')
plt.ylabel('Spending Score (1-100)')
plt.title('Spending Score (1-100) vs Annual Income (k$)')
plt.show()
```



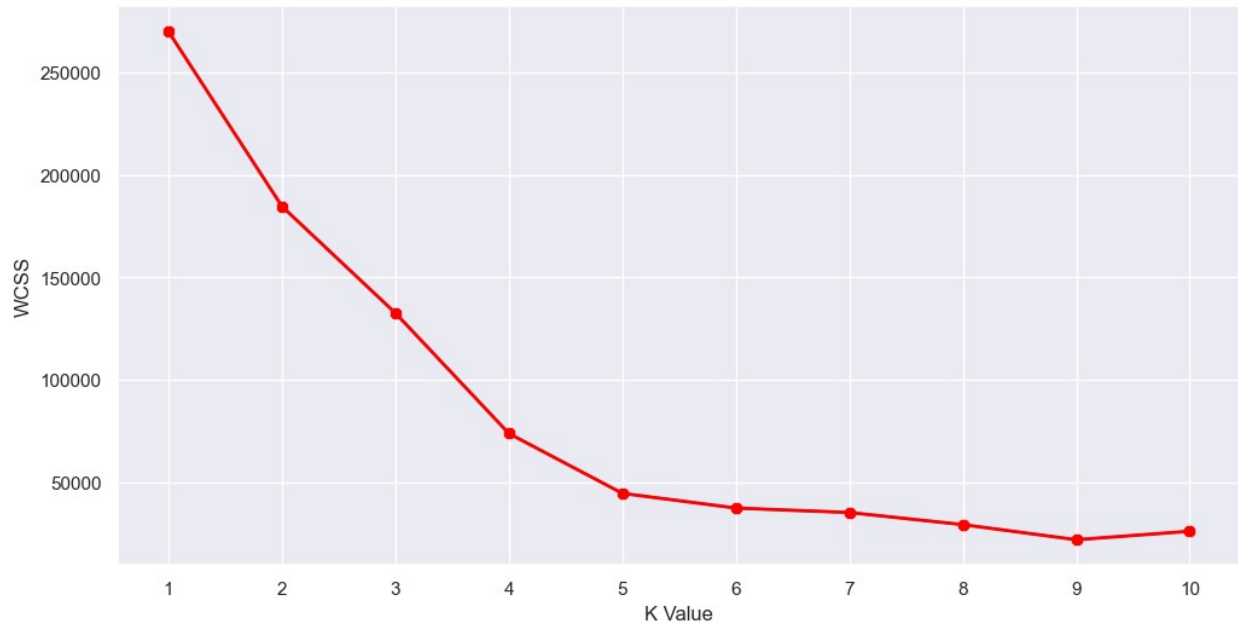
```
pip install scikit-learn
```

```
Defaulting to user installation because normal site-packages is not
writeable
Collecting scikit-learn
  Downloading scikit_learn-1.6.1-cp39-cp39-macosx_12_0_arm64.whl (11.1
MB)
acosx_12_0_arm64.whl (30.3 MB)
ent already satisfied: numpy>=1.19.5 in
/Users/jagdeepsingh/Library/Python/3.9/lib/python/site-packages (from
scikit-learn) (2.0.2)
Collecting joblib>=1.2.0
  Using cached joblib-1.4.2-py3-none-any.whl (301 kB)
Installing collected packages: threadpoolctl, scipy, joblib, scikit-
learn
Successfully installed joblib-1.4.2 scikit-learn-1.6.1 scipy-1.13.1
threadpoolctl-3.6.0
WARNING: You are using pip version 21.2.4; however, version 25.0.1 is
available.
You should consider upgrading via the
'/Library/Developer/CommandLineTools/usr/bin/python3 -m pip install --
upgrade pip' command.
Note: you may need to restart the kernel to use updated packages.
```

```
#Importing KMeans from sklearn
from sklearn.cluster import KMeans
```

```
wcss=[]
for i in range(1,11):
    km=KMeans(n_clusters=i)
    km.fit(X)
    wcss.append(km.inertia_)
```

```
#The elbow curve
plt.figure(figsize=(12,6))
plt.plot(range(1,11),wcss)
plt.plot(range(1,11),wcss, linewidth=2, color="red", marker ="8")
plt.xlabel("K Value")
plt.xticks(np.arange(1,11,1))
plt.ylabel("WCSS")
plt.show()
```



```
#Taking 5 clusters
kml=KMeans(n_clusters=5)
#Fitting the input data
kml.fit(X)
#predicting the labels of the input data
y=kml.predict(X)
#adding the labels to a column named label
df1["label"] = y
#The new dataframe with the clustering done
df1.head()
```

	CustomerID	Gender	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	0	19	15	39
1	2	0	21	15	81
2	3	1	20	16	6
3	4	1	23	16	77
4	5	1	31	17	40

```
#Scatterplot of the clusters
plt.figure(figsize=(10,6))
sns.scatterplot(x = 'Annual Income (k$)',y = 'Spending Score (1-100)',hue="label",
palette=['green','orange','brown','dodgerblue','red'],
```

```

legend='full',data = df1 ,s = 60 )
plt.xlabel('Annual Income (k$)')
plt.ylabel('Spending Score (1-100)')
plt.title('Spending Score (1-100) vs Annual Income (k$)')
plt.show()

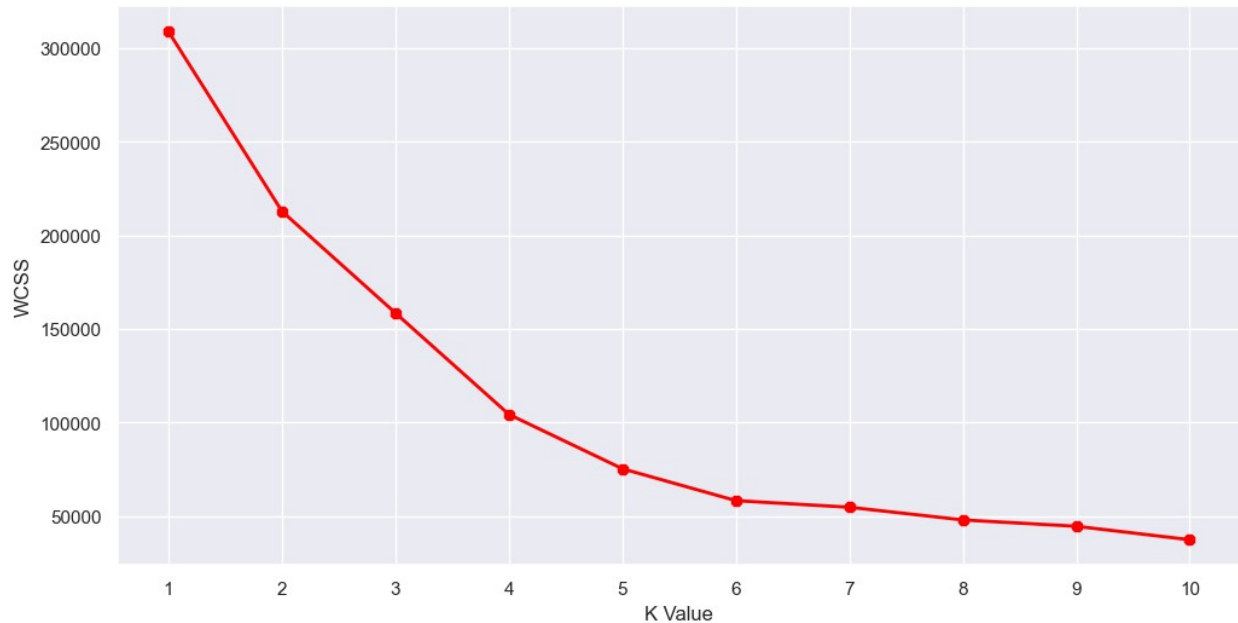
```



```

#Taking the features
X2=df1[["Age","Annual Income (k$)","Spending Score (1-100)"]]
#Now we calculate the Within Cluster Sum of Squared Errors (WSS) for
different values of k.
wcss = []
for k in range(1,11):
    kmeans = KMeans(n_clusters=k, init="k-means++")
    kmeans.fit(X2)
    wcss.append(kmeans.inertia_)
plt.figure(figsize=(12,6))
plt.plot(range(1,11),wcss, linewidth=2, color="red", marker ="8")
plt.xlabel("K Value")
plt.xticks(np.arange(1,11,1))
plt.ylabel("WCSS")
plt.show()

```



#We choose the k for which WSS starts to diminish

```
km2 = KMeans(n_clusters=5)
```

```
y2 = km.fit_predict(X2)
```

```
df1["label"] = y2
```

#The data with labels

```
df1.head()
```

	CustomerID	Gender	Age	Annual Income (k\$)	Spending Score (1-100)
label					
0	1	0	19	15	39
6					
1	2	0	21	15	81
5					
2	3	1	20	16	6
6					
3	4	1	23	16	77
5					
4	5	1	31	17	40
6					

#3D Plot as we did the clustering on the basis of 3 input features

```
fig = plt.figure(figsize=(20,10))
```

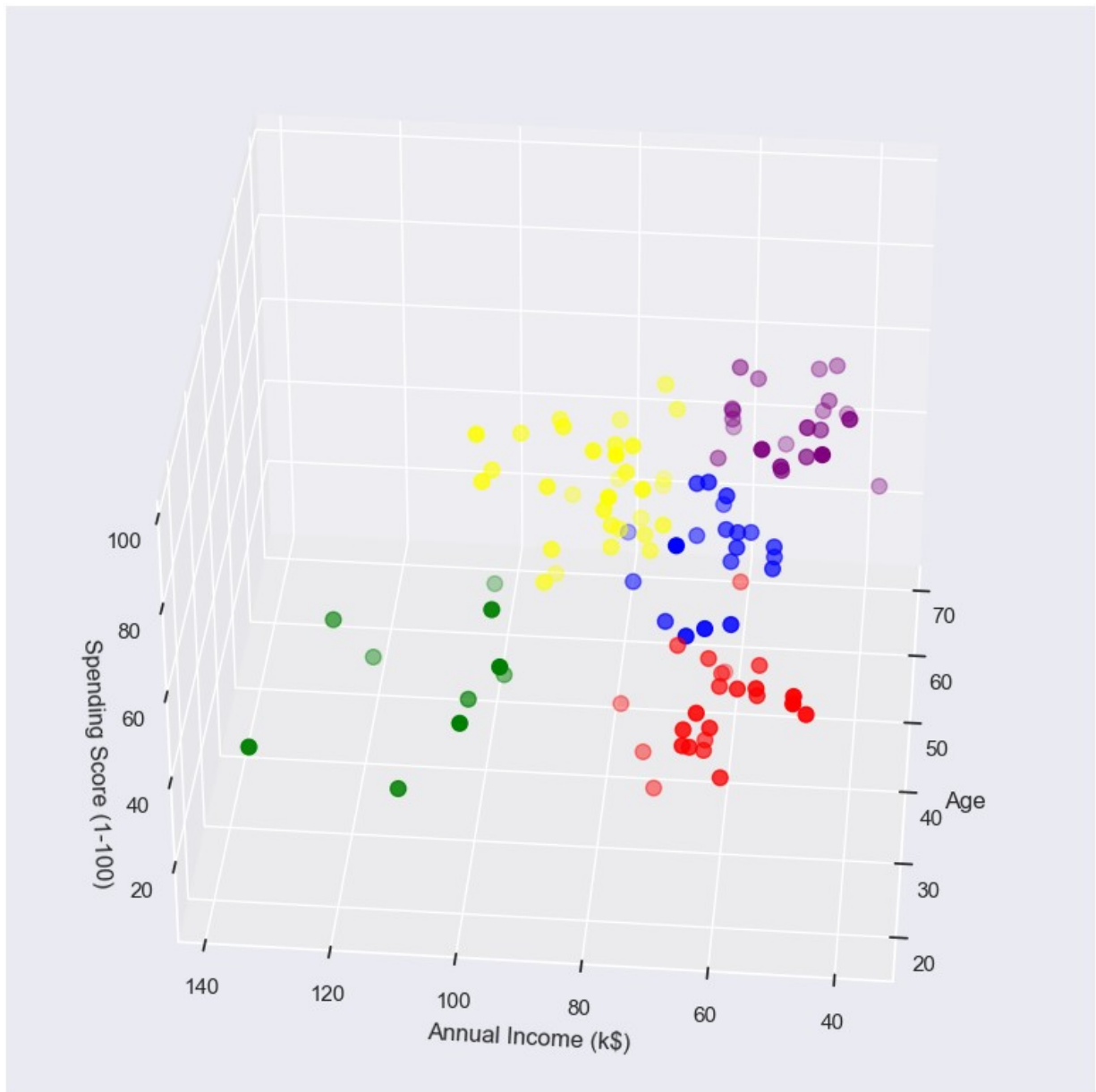
```
ax = fig.add_subplot(111, projection='3d')
```

```
ax.scatter(df1.Age[df1.label == 0], df1["Annual Income (k$)"]
[df1.label == 0], df1["Spending Score (1-100)"][df1.label == 0],
c='purple', s=60)
```

```
ax.scatter(df1.Age[df1.label == 1], df1["Annual Income (k$)"]
[df1.label == 1], df1["Spending Score (1-100)"][df1.label == 1],
c='red', s=60)
```

```
ax.scatter(df1.Age[df1.label == 2], df1["Annual Income (k$)"]
```

```
[df1.label == 2], df1["Spending Score (1-100)"][df1.label == 2],  
c='blue', s=60)  
ax.scatter(df1.Age[df1.label == 3], df1["Annual Income (k$)"]  
[df1.label == 3], df1["Spending Score (1-100)"][df1.label == 3],  
c='green', s=60)  
ax.scatter(df1.Age[df1.label == 4], df1["Annual Income (k$)"]  
[df1.label == 4], df1["Spending Score (1-100)"][df1.label == 4],  
c='yellow', s=60)  
ax.view_init(35, 185)  
plt.xlabel("Age")  
plt.ylabel("Annual Income (k$)")  
ax.set_zlabel('Spending Score (1-100)')  
plt.show()
```



```

cust1=df1[df1["label"]==1]
print('Number of customer in 1st group=', len(cust1))
print('They are -', cust1["CustomerID"].values)
print("-----")
cust2=df1[df1["label"]==2]
print('Number of customer in 2nd group=', len(cust2))
print('They are -', cust2["CustomerID"].values)
print("-----")
cust3=df1[df1["label"]==0]
print('Number of customer in 3rd group=', len(cust3))
print('They are -', cust3["CustomerID"].values)
print("-----")

```

```

cust4=df1[df1["label"]==3]
print('Number of customer in 4th group=', len(cust4))
print('They are -', cust4["CustomerID"].values)
print("-----")
cust5=df1[df1["label"]==4]
print('Number of customer in 5th group=', len(cust5))
print('They are -', cust5["CustomerID"].values)
print("-----")

Number of customer in 1st group= 24
They are - [ 62  66  69  76  79  85  88  89  92  95  96  98 100 101
104 106 112 114
115 116 121 125 133 143]
-----

Number of customer in 2nd group= 20
They are - [ 80  84  86  90  93  94  97  99 102 105 108 113 118 119
120 122 123 127
147 161]
-----

Number of customer in 3rd group= 22
They are - [ 41  54  58  61  63  64  65  68  71  73  74  75  81  83
87  91 103 107
109 110 111 117]
-----

Number of customer in 4th group= 10
They are - [181 183 185 187 189 191 193 195 197 199]
-----

Number of customer in 5th group= 32
They are - [124 126 128 130 132 134 136 138 140 142 144 146 148 150
152 154 156 158
160 162 164 166 168 170 172 174 176 178 180 182 184 186]
-----

```